

Renu Arya

Cognitive Science Researcher

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Google Scholar | GitHub | LinkedIn

Education

M.Sc. Cognitive Science | Indian Institute of Technology Delhi | 2023–2025

B.A. Psychology | Kurukshetra University | 2018–2021

Senior Secondary (PCMB) | 82.8% | Kurukshetra University | 2016

Secondary (Class X) | 96% | CBSE | 2014

Research Experience

Research Associate | Marbles Health | Oct 2025 – Present

Investigating neurological impacts of non-invasive brain stimulation (tDCS) across psychiatric disorders (ADHD, OCD, Depression).

Researcher | NCNL, SAIDE, IIT Jodhpur | May–Sept 2025

Processed 200+ fMRI datasets for network neuroscience analysis pipeline.

Graduate Researcher | LINCS Group, IIT Delhi | Aug 2024 – May 2025

Modeled 8,000+ behavioral trials using Drift Diffusion Models.

Research Assistant | TRANSiT Lab, IIT Kanpur | May 2024 – Mar 2025

Signal processing analysis of tRNS treatment in ADHD populations.

Publications

Dakwar-Kawar et al. (2025). Transcranial random noise stimulation on excitation/inhibition balance in ADHD. *NeuroImage: Clinical*, 49, 103923.

Research Projects

Pilot Mental Health Survey | IIT Kanpur | May–Jun 2025

Co-led analysis of mental health survey (N=50) among IITK students.

n-back Working Memory Task | IIT Delhi | Oct 2024

Designed 2-back task using MATLAB/PsychToolbox, analyzed cognitive load effects.

Fatigue & Cognitive Performance | IIT Delhi | Jan–Mar 2025

Longitudinal study demonstrating rest breaks improved RT (98.2s to 81.1s).

Technical Skills

Neuroimaging: SPM, FSL, EEGLAB, Brainstorm

Methods: DDM, Bayesian Modeling, Time-series Analysis

Programming: Python (pandas, NumPy), R, SQL, MATLAB

Tools: Git, Jupyter, VS Code, PsychToolbox, Tableau